# Patterns In Youth Marijuana Use

*Understanding Drug Usage in Minors through Decision Trees and Ensemble Methods*

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# Introduction

Marijuana is the most commonly used illicit drug by US adolescents. While the usage rates have remained stable over time in youth as legislation around marijuana has changed, the potential for more negative health outcomes still persist. [1]

This project seeks to investigate which demographic and environmental factors are most related to the usage of marijuana in US teens under the age of 18.

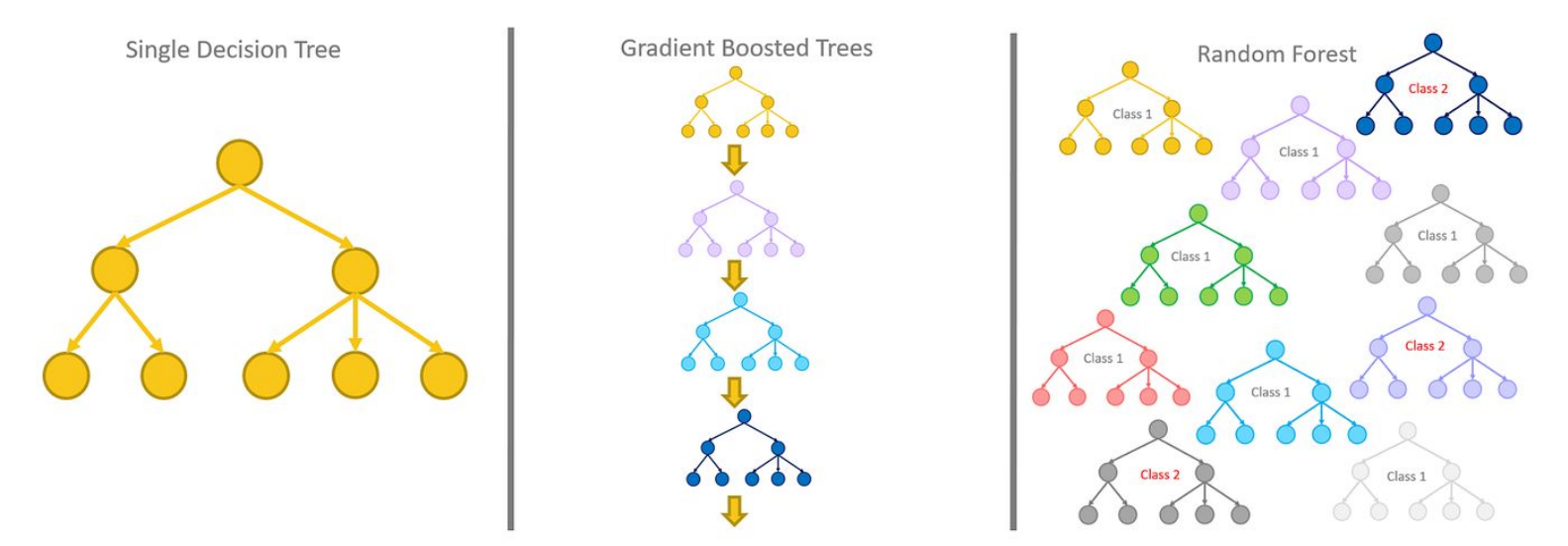
# Introduction: Key Goals and Approach

- Can decision trees and ensemble methods (bootstrap aggregating, random forests, gradient boosting) be used to predict marijuana use in youth?
- Three approaches: **binary classification** (has used before vs. has not), **multi-class classification** (frequency of usage in the past year), **regression** (at what age did an individual begin using)
- Which approach and model generate the highest accuracy, and which features most strongly associate with youth cannabis use?

# Introduction: Dataset

- National Survey on Drug Use and Health (2023) from Substance Abuse and Mental Health Services Administration
- +10,000 observations (respondents) across 79 features (survey questions)
- Features/predictors of interest: demography, parental influence, parental attitudes on drug usage

# Theoretical Background



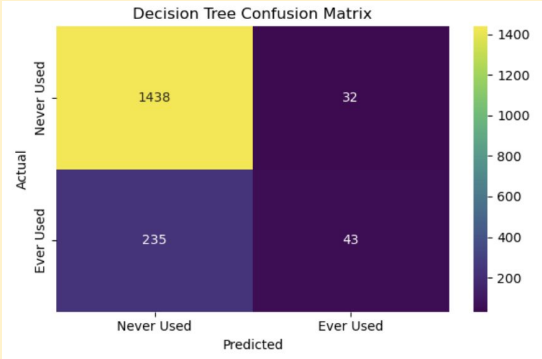

Silipo, R. (2020, March 19). *Ensemble models: Bagging & boosting*. Medium. https://medium.com/analytics-vidhya/ensemble-models-bagging-boosting-c33706db0b0b [5]

# Data Methodology

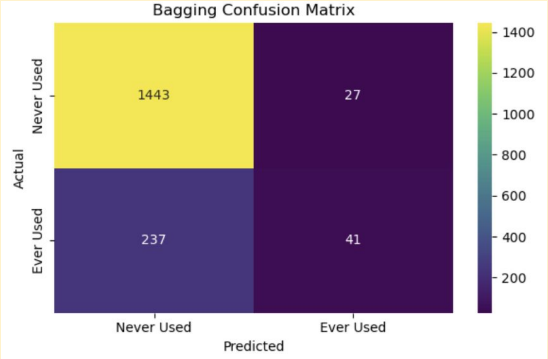
- Data filtered to include only responses for those under 18 (~30,000 observations to ~10,000)
- Variables narrowed down to 14 total predictors
- Three target variables of interest:
  - 'MRJFLAG': has or has not used marijuana
  - 'IRMJFY': frequency of use of marijuana in past year (generalized into three classes)
  - 'IRMJAGE': age of first use of marijuana (excluding all respondents who have never used marijuana)
- Data split using 80/20 train-test split, implemented all models and training data and tested for accuracy with new data
- Classification assessed using accuracy and confusion matrices; regression assessed using metrics such as MSE, RMSE, and $R^2$
- Tools utilized in Python included NumPy [2], Matplotlib [3], scikit-learn [4], Pandas [6], and Seaborn [7]

# Binary Classification Results

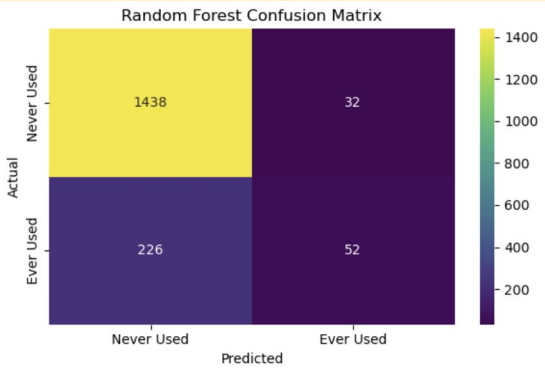
## Decision Tree (accuracy: 84.73%)



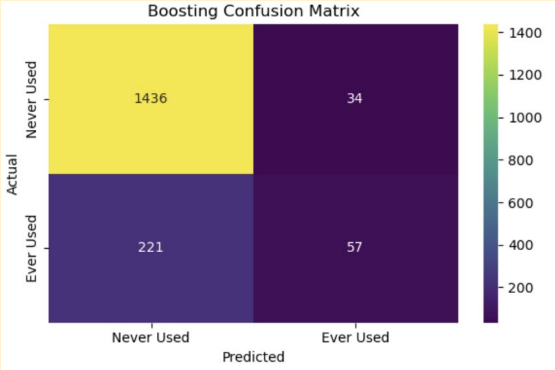

## Bagging (accuracy: 84.90%)




## Random Forest (accuracy: 85.24%)

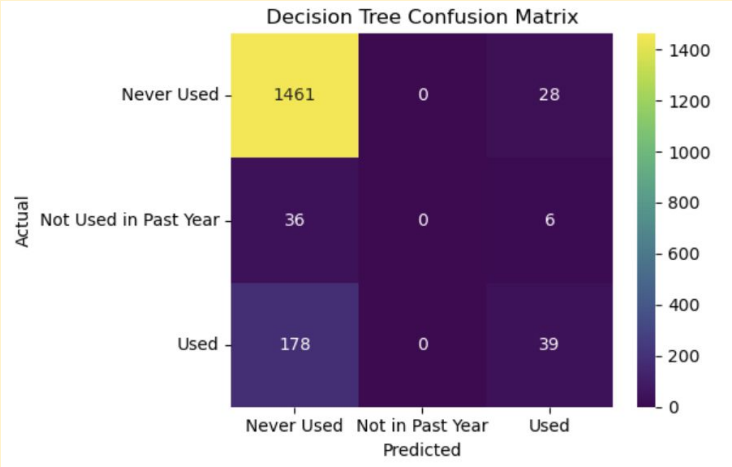



## Boosting (accuracy: 85.41%)

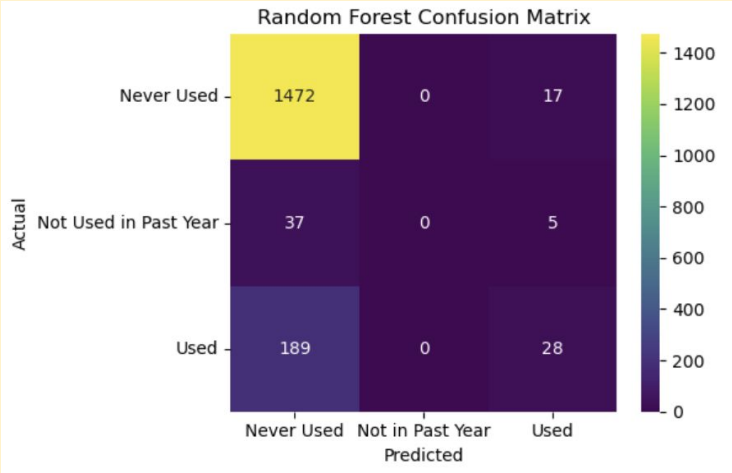


# Multi-class Classification Results

Decision Tree (accuracy: 85.81%)




Random Forest (accuracy: 85.81%)



# Regression Results

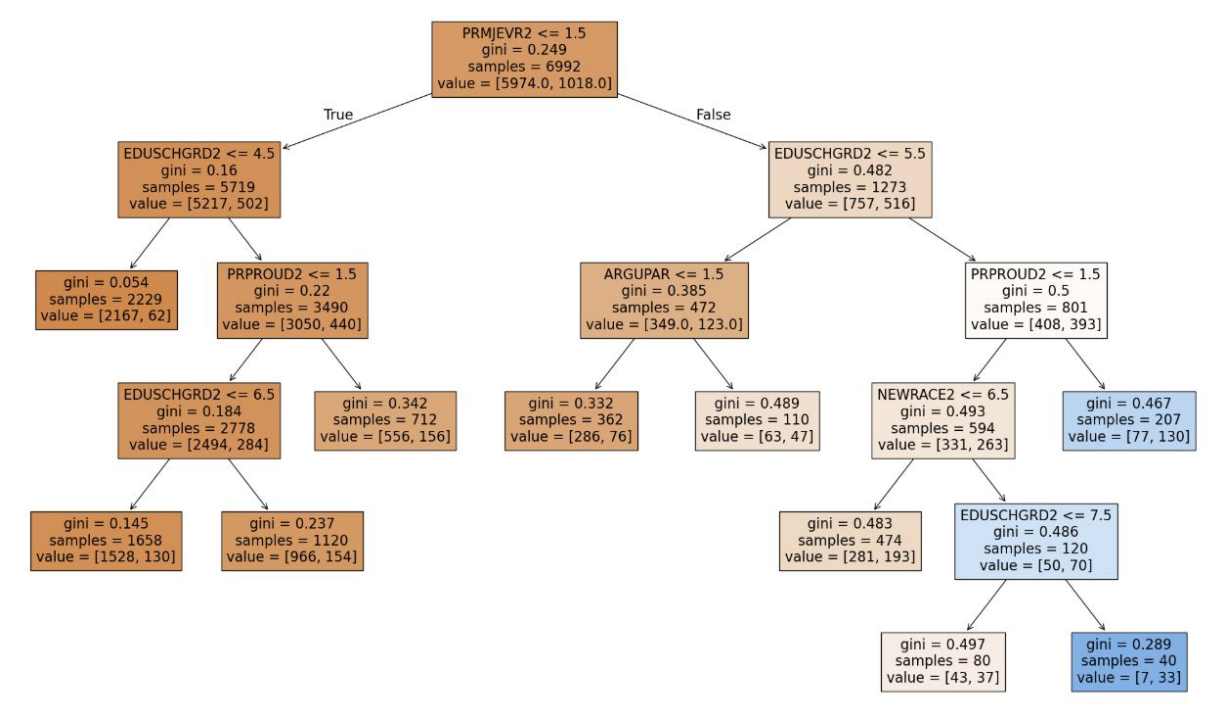
| Model | MSE | RMSE | $R^2$ |
|---|---|---|---|
| **Decision Tree** | 2.5586 | 1.5996 | 0.1173 |
| **Boosting ($\lambda$ = 0.01)** | 2.4854 | 1.5765 | 0.1425 |

- Standard deviation of age of first use = 1.74
- Boosting outperforms pruned decision tree

# Discussion

- **Binary classification:** boosting model achieved highest accuracy (85.41%) and detected the highest proportion of those who had used marijuana correctly; random forest was second best (85.24% accuracy)
- **Multi-class classification:** both models achieved the same accuracy (85.81%), but decision tree detected highest proportion of those who had used in the past year; both models struggled with minority class
- **Regression:** boosting model with optimal shrinkage parameter achieved highest accuracy (MSE: ~2.49)

# Discussion: Pruned Decision Tree



# Discussion: Feature Importance

| | feature_name | importance |
|---|---|---|
| 0 | PRMJEVR2 | 0.458557 |
| 1 | EDUSCHGRD2 | 0.195994 |
| 2 | PRPROUD2 | 0.059855 |
| 3 | ARGUPAR | 0.048910 |
| 4 | NEWRACE2 | 0.042318 |
| 5 | INCOME | 0.034213 |
| 6 | PRLMTTV2 | 0.032433 |
| 7 | COUTYP4 | 0.031819 |
| 8 | PARLMTSN | 0.023744 |
| 9 | IRSEX | 0.023355 |
| 10 | PARCHKHW | 0.013720 |
| 11 | RLGATTD | 0.013384 |
| 12 | PARHLPHW | 0.012720 |
| 13 | PRVDRGO2 | 0.008978 |

Binary Classification

| | feature_name | importance |
|---|---|---|
| 0 | PRMJEVR2 | 0.668774 |
| 1 | EDUSCHGRD2 | 0.210870 |
| 2 | ARGUPAR | 0.055392 |
| 3 | PRPROUD2 | 0.048322 |
| 4 | NEWRACE2 | 0.016643 |
| 5 | IRSEX | 0.000000 |
| 6 | INCOME | 0.000000 |
| 7 | COUTYP4 | 0.000000 |
| 8 | RLGATTD | 0.000000 |
| 9 | PRVDRGO2 | 0.000000 |
| 10 | PARCHKHW | 0.000000 |
| 11 | PARHLPHW | 0.000000 |
| 12 | PRLMTTV2 | 0.000000 |
| 13 | PARLMTSN | 0.000000 |

Multi-class Classification

| | feature_name | importance |
|---|---|---|
| 0 | EDUSCHGRD2 | 0.528574 |
| 1 | COUTYP4 | 0.081930 |
| 2 | INCOME | 0.072122 |
| 3 | NEWRACE2 | 0.071223 |
| 4 | PRMJEVR2 | 0.048364 |
| 5 | IRSEX | 0.033098 |
| 6 | ARGUPAR | 0.032341 |
| 7 | PARCHKHW | 0.028665 |
| 8 | PARLMTSN | 0.027232 |
| 9 | PRPROUD2 | 0.025352 |
| 10 | RLGATTD | 0.018760 |
| 11 | PARHLPHW | 0.014642 |
| 12 | PRVDRGO2 | 0.011350 |
| 13 | PRLMTTV2 | 0.006348 |

Regression

# Discussion: Potential Limitations

- Class imbalance in both formulations of classification
- Grade level is potentially redundant in the regression problem
- Youth responses to the survey may be inaccurate
- Broad amount of factors that could also contribute to youth marijuana use

# Conclusion

- Parental attitudes towards drug use appear to have the strongest assocation with US youth usage
- Grade level, race, and income also demonstrate association
- Ensemble methods offer more thorough predictive capability
- Future research built on these methods to deal with class imbalance

# Bibliography

[1] Hammond, C. J., Chaney, A., Hendrickson, B., & Sharma, P. (2020). Cannabis use among U.S. adolescents in the era of marijuana legalization: a review of changing use patterns, comorbidity, and health correlates. *International review of psychiatry (Abingdon, England)*, *32*(3), 221–234. https://doi.org/10.1080/09540261.2020.1713056

[2] Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., … & Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, *585*, 357-362.

[3] Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. *Computing in Science & Engineering*, *9*(3), 90-95.

[4] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., … & Duchesneau, E. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, *12*, 2825-2830.

[5] Silipo, R. (2020, March 19). Ensemble models: Bagging & boosting. *Medium*. https://medium.com/analytics-vidhya/ensemble-models-bagging-boosting-c33706db0b0b

[6] The pandas development team. (2020). *pandas-dev/pandas: Pandas*. Zenodo. https://doi.org/10.5281/zenodo.3509134

[7] Waskom, M. L. (2021). Seaborn: Statistical Data Visualization. *Journal of Open Source Software*, *6*(60), 3021.